Paper IV

Verification closeout

Cyclic analytic functions, reciprocal families, and spectral zero data

Version 1.1.0 | 19 October 2026 (UTC)

Result

The existing 68-result formal core passed a fresh pinned-source replay, declaration-kind resolution, transitive axiom audit, and compiler controls. The mathematical sources and their hypotheses were not changed. This report closes the verification task, rather than introducing a new mathematical theorem.

Gate	Verified scope
Production compilation	118 of 118 modules, with new project objects
Mathematical source preservation	All 119 Lean source files unchanged
Reviewed target kinds	1,164 targets: 953 theorems and 211 constructions
Namespace axiom audit	2,234 declarations; 1,999 theorem declarations
Real compiler controls	Six accepted outcomes under explicit diagnostics
Verifier and distribution tests	42 verification tests; 10 integrity tests

What completion means here

The 151-row reviewed map covers the full scoped statements of the 68 theorem-like manuscript records. Declaration counts include supporting and generated results and must not be read as independent discovery counts. The statement correspondence is inherited and checked against its exact contract, not newly re-proved line by line in this pass.

Source and proof evidence

A new project build, not a reused project cache

The replay began with an empty project-object directory. Each production module was compiled separately with one Lean worker. Every successful source and output object is fingerprinted in the cumulative receipt. The aggregate imports the completed production closure. The separate original audit source and the generated reviewed-kind consumer were then executed.

The compiler was checked by exact binary hash and commit. All nine external package Git revisions match the project dependency manifest. Cached external objects were reused. This is not a cold rebuild of Mathlib or an independently implemented proof kernel.

Pin	Value
Lean	4.34.0-rc2
Compiler commit	6a10ac8c22beadecabdbb0919c2b50214762f91d
Mathlib revision	aeb72f1e9981b6d35b426b16dca7759be3e7ba06
Permitted transitive axioms	propext, Classical.choice, Quot.sound

Compiler fingerprint

```
e8baaa71855a616dc351028f3ad2200051b0671f423a1696a100e809302d5550
```

Control contract

A valid projector-idempotence application is accepted. An explicit false theorem, an admitted proof, an omitted premise, an injected custom axiom, and a definition presented as a theorem must produce their intended rejection diagnostics. A timeout, signal kill, missing import or unrelated API error does not count as successful rejection.

Evidence files

```
verification/lean_primary/REPLAY.json
verification/lean_primary/AXIOM_REPORTS.json
verification/SOURCE_MAP_CHECK.json
verification/SOURCE_CONTRACT.json
```

Full compiler logs, generated consumers, fixtures, source hashes and the project output objects are retained. The external runtime and dependency cache are not distributed in this package.

Finite verification and controls

The finite computations were executed separately from Lean. Their domains overlap, so the rows below are not added into a single inflated total. These tests support the implementation and examples. Universal statements are supported by formal proof terms under their hypotheses.

Suite	Actual result
Exact algebra	1,008 exact rational assertions
Visual examples	147 symbolic, finite and enclosure checks
Original tables	981 of 981 rows checked
Strengthened-paper suite	1,246 of 1,246 checks passed
Original-paper suite	647 of 647 checks passed
Rational subset and bracket records	818 rational table rows; four bracket records

Verification software itself was tested

The 42-test suite uses the recovered typed source contract and adversarial mutations of rows, targets, evidence labels, axiom reports and process outcomes. The ten integrity tests independently cover changed bytes, missing files, unexpected files, duplicate records, unsafe paths, symlinks and prohibited font files, together with positive and empty-manifest cases.

Observed computation environment

Python 3.13.5. sympy 1.14.0, mpmath 1.3.0, numpy 2.3.5, pandas 2.2.3, matplotlib 3.10.8. Versions are detected at execution time, not copied from the older availability report.

The historical 836-file baseline remained byte-identical before and after the finite replay. The old result files were not overwritten. New logs and results are stored in a separate verification directory.

```
verification/finite_primary/FINITE_REPLAY.json
verification/VERIFIER_TESTS.json
verification/INTEGRITY_TESTS.json
```

Failures and repairs

Warning policy

The inherited `-E warning` setting accepted an isolated deliberately admitted proof with exit code 0. The corrected `-DwarningAsError=true` setting rejected the same fixture with exit code 1. The active replay uses the corrected policy. The production source files were not weakened or edited.

Startup race in a control fixture

The first new verifier suite passed 37 of 38 tests. Its descendant-cleanup test timed out before the parent announced startup, so it did not establish that a descendant had started. The repaired fixture waits for explicit child readiness, then applies a separate bounded execution deadline. The final 42-test suite passes. The failed first run remains in the archive.

Distribution and availability metadata

The old 908-row distribution manifest still listed 73 excluded font files. Its verifier therefore failed. The recovered 836-file tree is preserved and a new top-level manifest records the actual distribution. No fonts were restored. The older available-check runner also described installed symbolic tools as unavailable. The new runner reports actual dependencies and executes the underlying suites.

What could not be recovered

The exact earlier unsaved fixture-control failure was not recovered. This release does not claim to reproduce that absent script. Instead, it directly exercises the recovered checker API using its actual typed records, and records new compilation, audit and control evidence. An initial network-based runtime recovery was unavailable in this container. The archived offline runtime supplied the successful pinned compiler and cache.

Failures are preserved as failures. Successful replacement checks have their own source fingerprints and receipts. No licence was selected or changed.

Reproduction and remaining scope

Three reproducibility levels

```
python3 -B scripts/verify_release.py
python3 -B scripts/run_checks.py --mode finite --work /tmp/paper-iv-finite
bash Run-Replay.sh /path/to/Komorebi_Portable_Lean_4.34.0-rc2 /tmp/paper-iv-full
```

Use a new, writable working directory outside the frozen release. The first command verifies stored evidence and file integrity, not the kernel again. Finite mode reruns the examples and verifier controls. Full mode additionally compiles all production modules and repeats the formal audits. No command downloads or installs dependencies.

The complete replay targets Linux or WSL with Python 3.11 or newer, the pinned Lean runtime, and matching cached packages. The actual run used a four-GiB container, one Lean worker, a 3,072-MiB Lean heap setting and a 300-second per-command deadline. This is not a native-Windows or cross-platform reproducibility claim.

Reader continuity

The 77-page monograph, 28-page visual guide and 140-page Lean guide are preserved dated readers, totaling 245 pages. This new report supplies current execution status. Earlier verification descriptions retain their historical dates. The latest visual reader was recovered separately, including its branch-sensitive clarification on reciprocal powers. Available editable sources and their baseline provenance are retained.

Mathematical boundaries

The declared core does not assert universal simplicity of all zeros, positivity of every reduced zero, a full displacement expansion or a zeta functional equation. Eventual simplicity, branch choices, multiplicities and nearest-pole asymptotics retain their stated hypotheses and ranges. Imported standard fact 11.1 remains outside the 68-result contract.

Next restart point

Preserve this release and its separate final delivery receipt. Paper IV no longer needs a fresh core replay merely because its older reader describes September evidence. The next twelve-paper verification task is Paper X. No status of another paper was upgraded by this closeout.